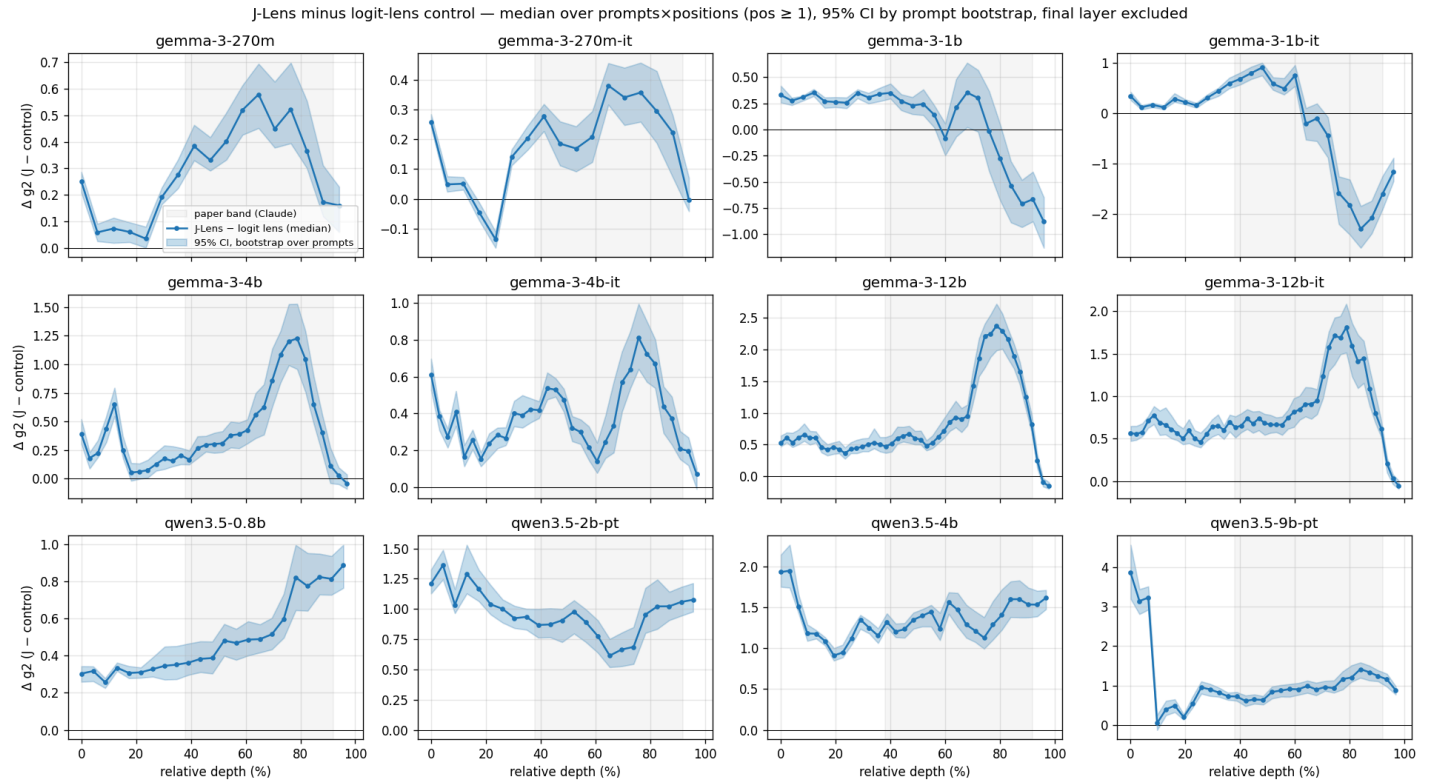


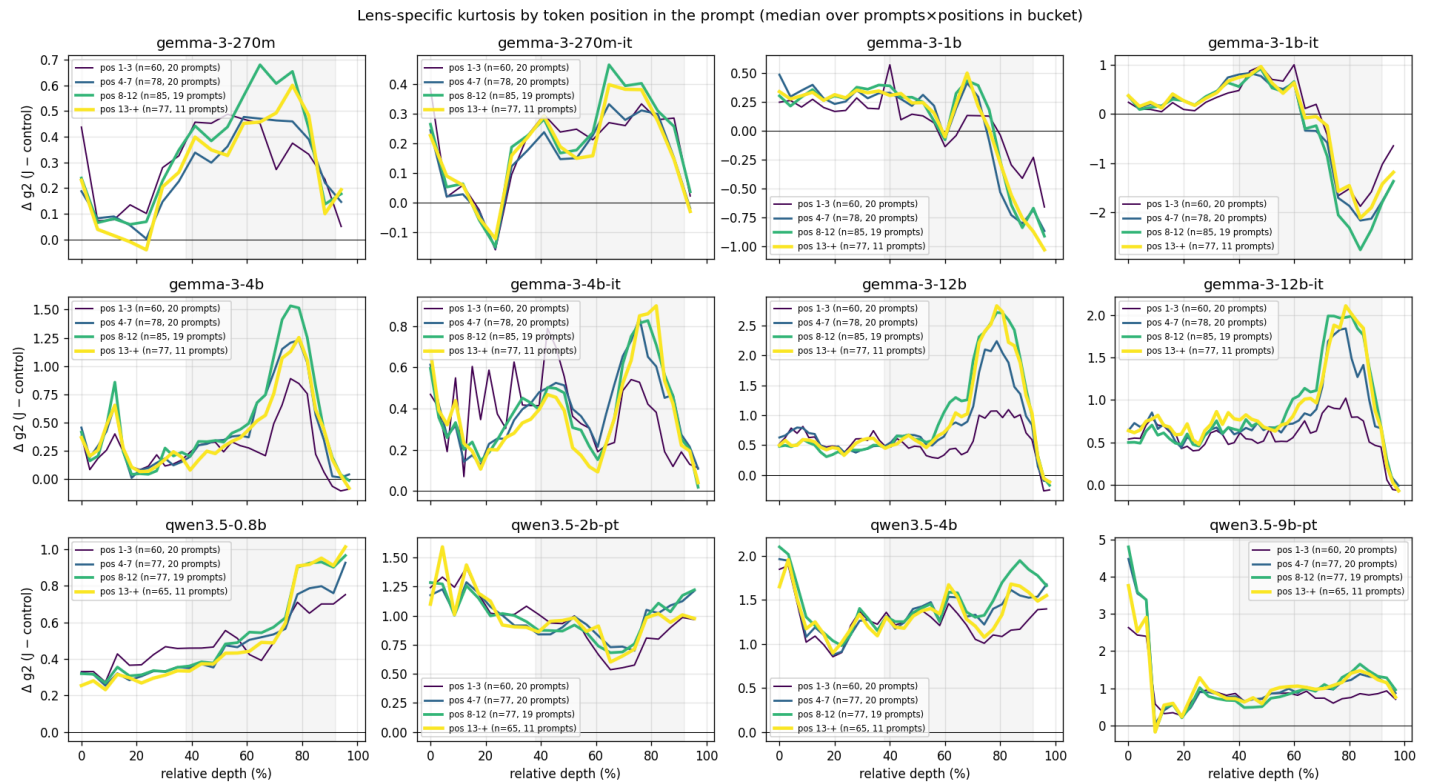
Kurtosis sweep — reading report

20 prompts, positions ≥ 1 (BOS excluded), final layer excluded, biased g2 estimator in float64. J-C = median J-Lens minus median logit-lens; CI95 = bootstrap over the 20 prompts; “bump” = peak of J-C over 55-95 % depth minus baseline (median over 25-55 %). Grey band = the paper’s workspace band (Claude, 38-92 %). Generated 3 Sept. 2026.

Overview — J-Lens minus control, all models



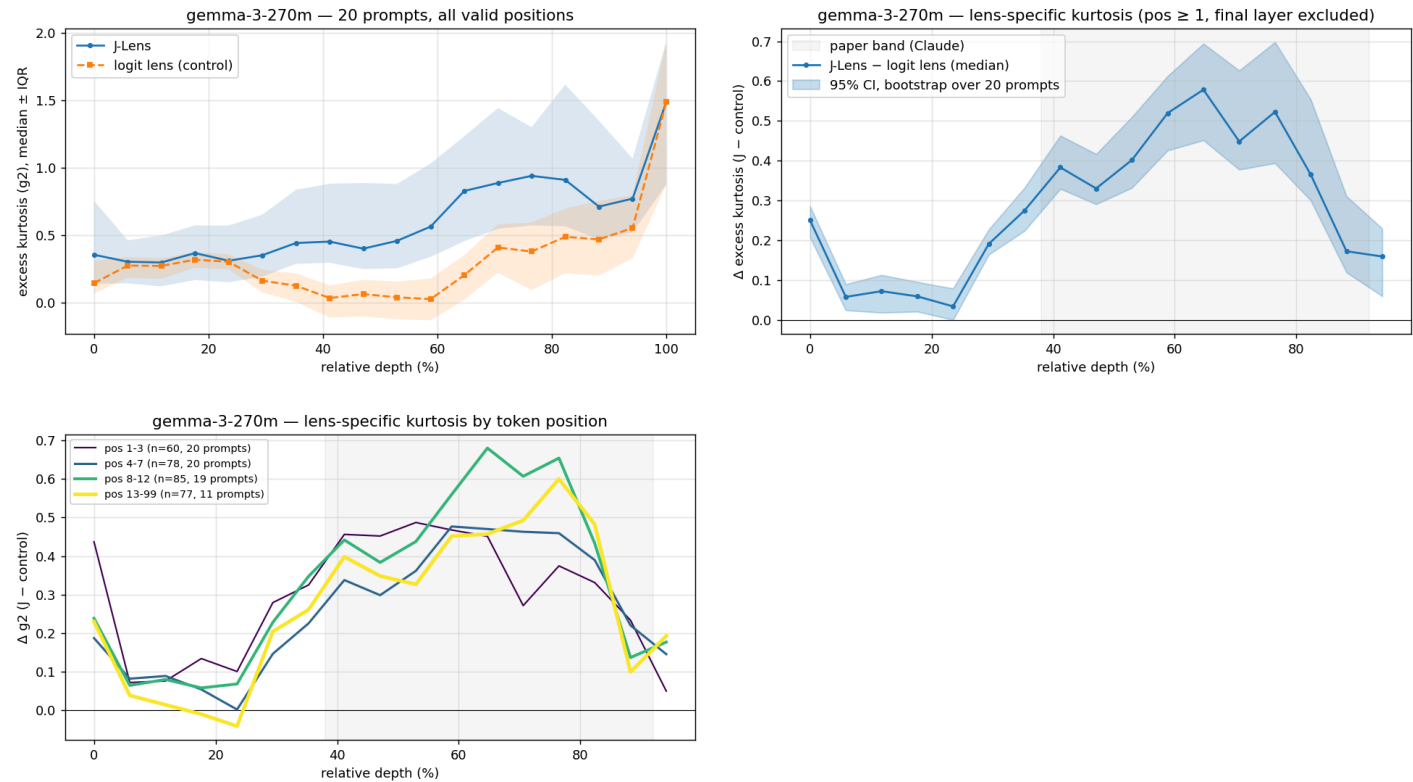
Overview — by position in the prompt



Family gemma-3

gemma-3-270m18 layers · d_model 640 · float32

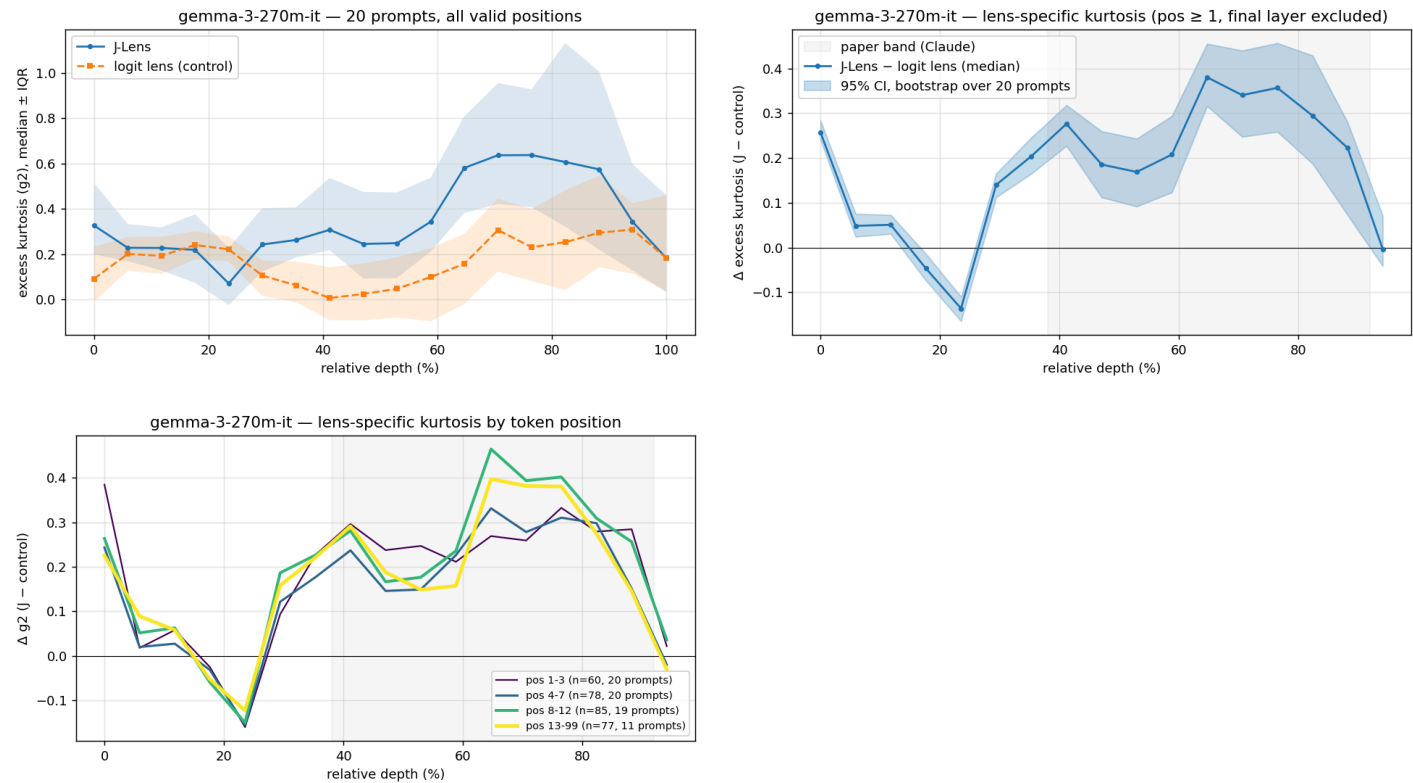
J-C baseline	J-C peak	bump	CI95 (prompts)	prompts > 0	onset	pos 1-5 → 11-15
0.33	0.58 @ 65 %	0.25	[0.14 ; 0.37]	20/20	—	0.43 → 0.80 (7/10)



gemma-3-270m-it

18 layers · d_model 640 · float32

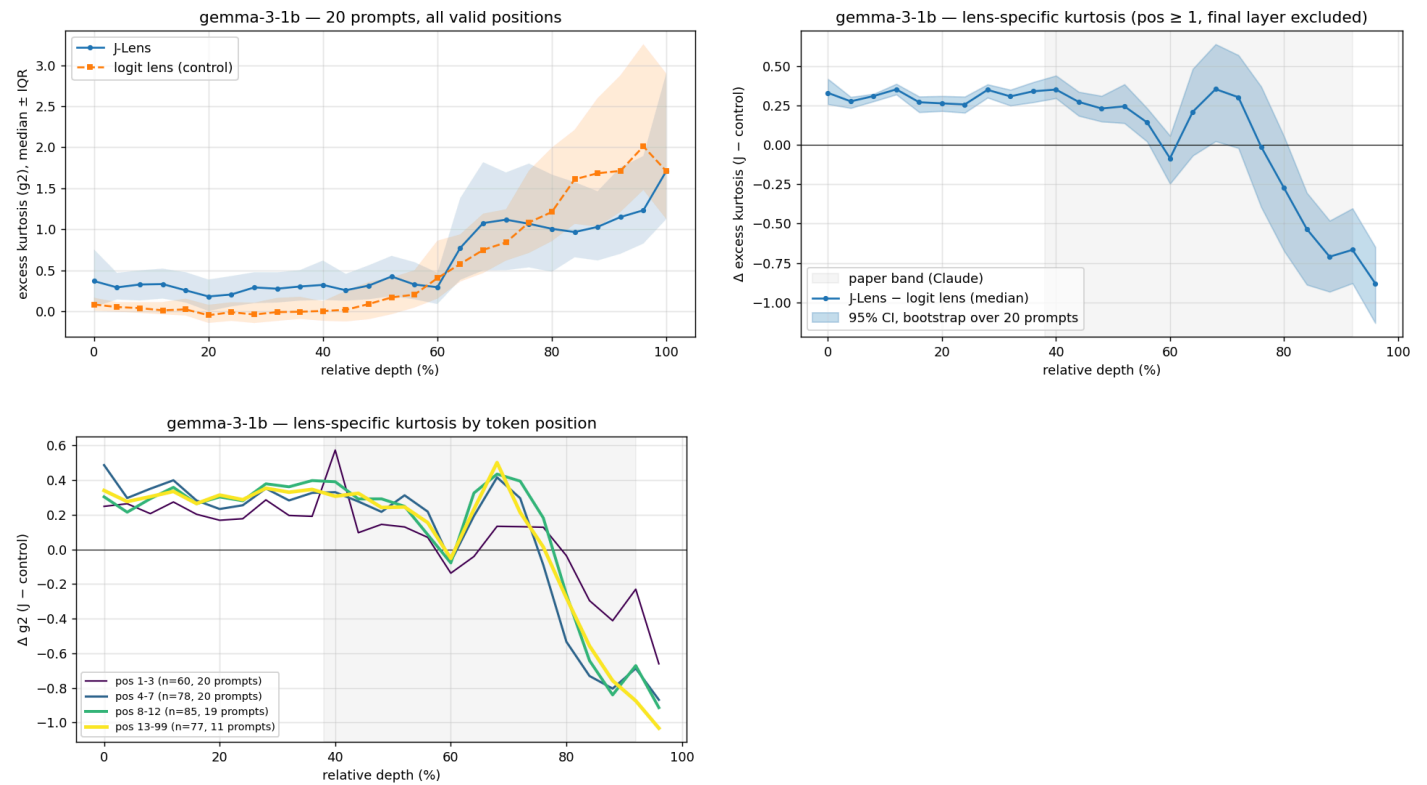
J-C baseline	J-C peak	bump	CI95 (prompts)	prompts > 0	onset	pos 1-5 → 11-15
0.19	0.38 @ 65 %	0.19	[0.12 ; 0.30]	20/20	—	0.27 → 0.57 (9/10)



gemma-3-1b 26 layers · d_model 1152 · bfloat16

J-C baseline	J-C peak	bump	CI95 (prompts)	prompts > 0	onset	pos 1-5 → 11-15
0.31	0.35 @ 68 %	0.05	[−0.23 ; 0.30]	13/20	—	0.12 → 0.50 (7/10)

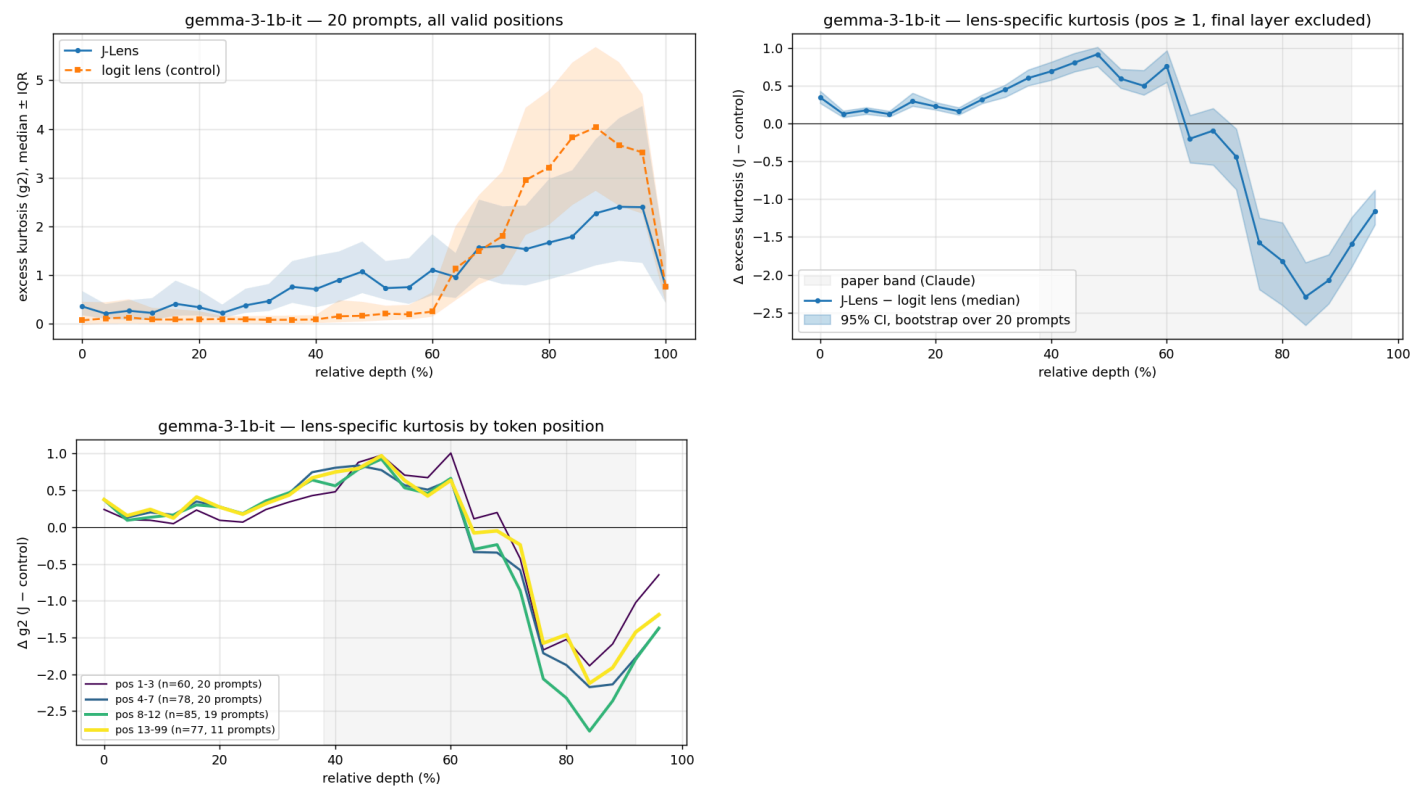
J-C goes below zero inside the band (min −0.71) · CI95 of the bump contains zero · control sharper than J-Lens at the end of the network (2.14 vs 1.26 at the penultimate layer)



gemma-3-1b-it 26 layers · d_model 1152 · bfloat16

J-C baseline	J-C peak	bump	CI95 (prompts)	prompts > 0	onset	pos 1-5 → 11-15
0.60	0.75 @ 60 %	0.16	[−0.00 ; 0.29]	12/20	48 %	0.92 → 0.98 (5/10)

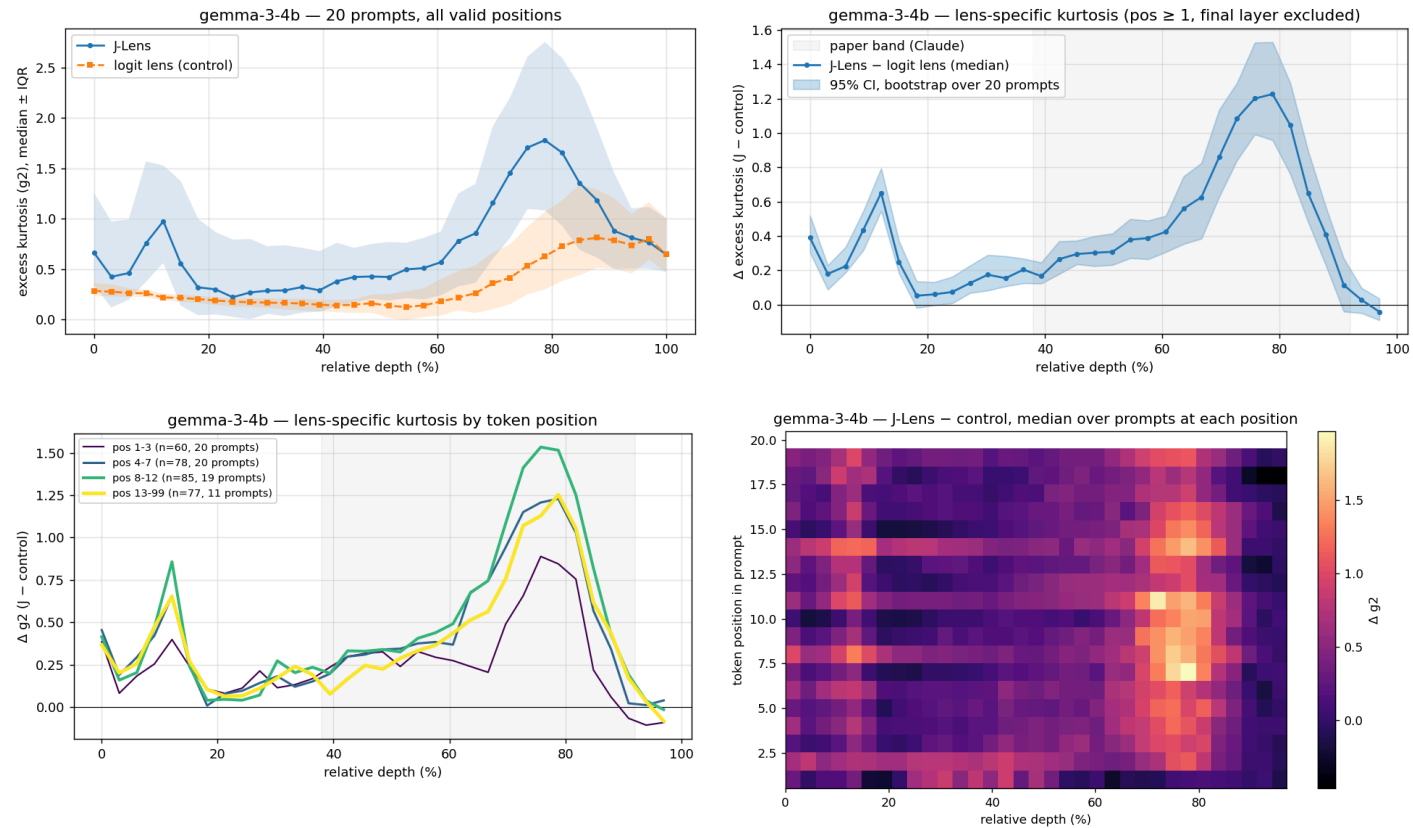
J-C goes below zero inside the band (min −2.29) · CI95 of the bump contains zero · control sharper than J-Lens at the end of the network (3.40 vs 2.24 at the penultimate layer)



gemma-3-4b
34 layers · d_model 2560 · bfloat16

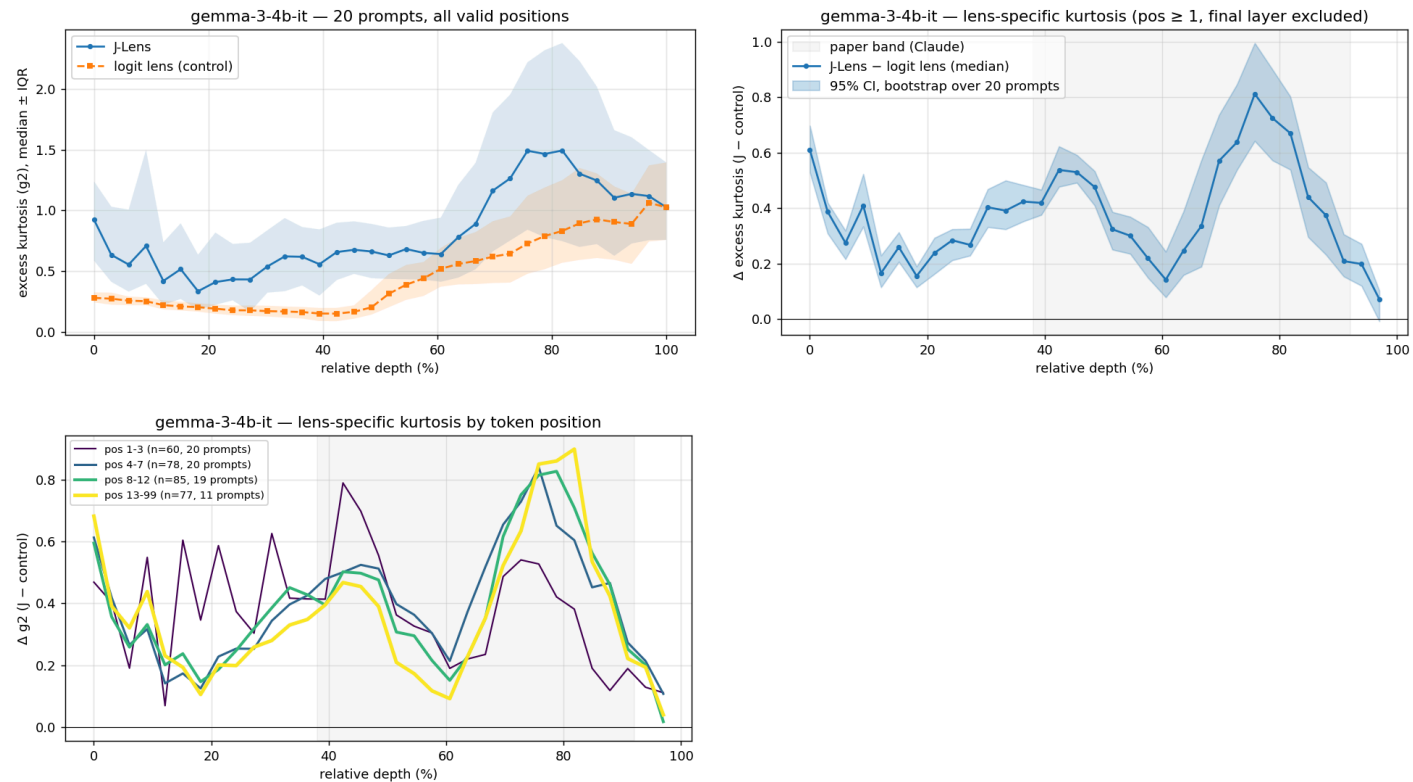
J-C baseline	J-C peak	bump	CI95 (prompts)	prompts > 0	onset	pos 1-5 → 11-15
0.23	1.23 @ 79 %	0.99	[0.79 ; 1.27]	20/20	64 %	0.92 → 1.34 (8/10)

clear position effect inside the same prompts



gemma-3-4b-it
34 layers · d_model 2560 · bfloat16

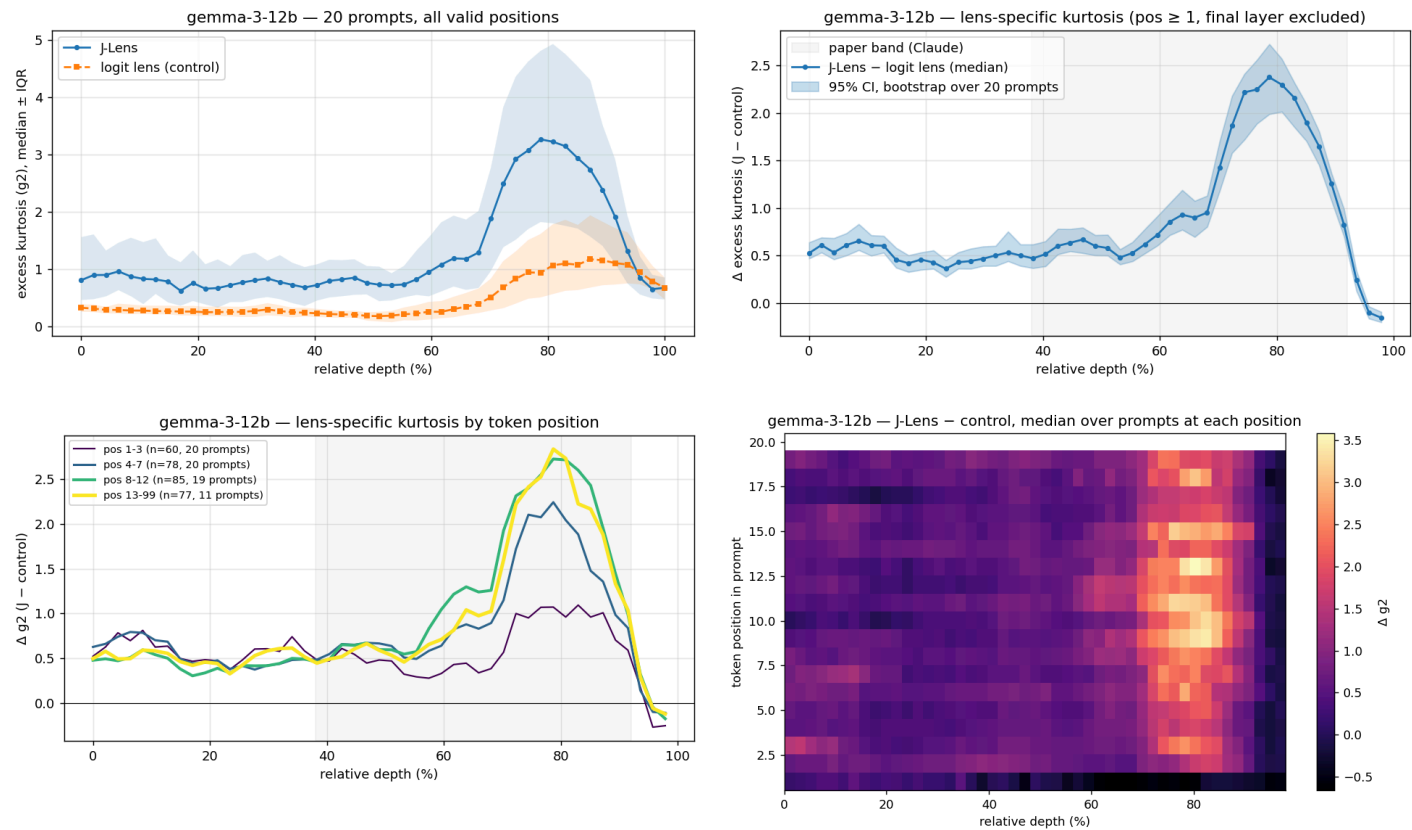
J-C baseline	J-C peak	bump	CI95 (prompts)	prompts > 0	onset	pos 1-5 → 11-15
0.41	0.81 @ 76 %	0.40	[0.24 ; 0.55]	19/20	76 %	0.51 → 1.05 (7/10)



gemma-3-12b 48 layers · d_model 3840 · bfloat16

J-C baseline	J-C peak	bump	CI95 (prompts)	prompts > 0	onset	pos 1-5 → 11-15
0.51	2.37 @ 79 %	1.87	[1.58 ; 2.17]	20/20	62 %	1.51 → 2.64 (9/10)

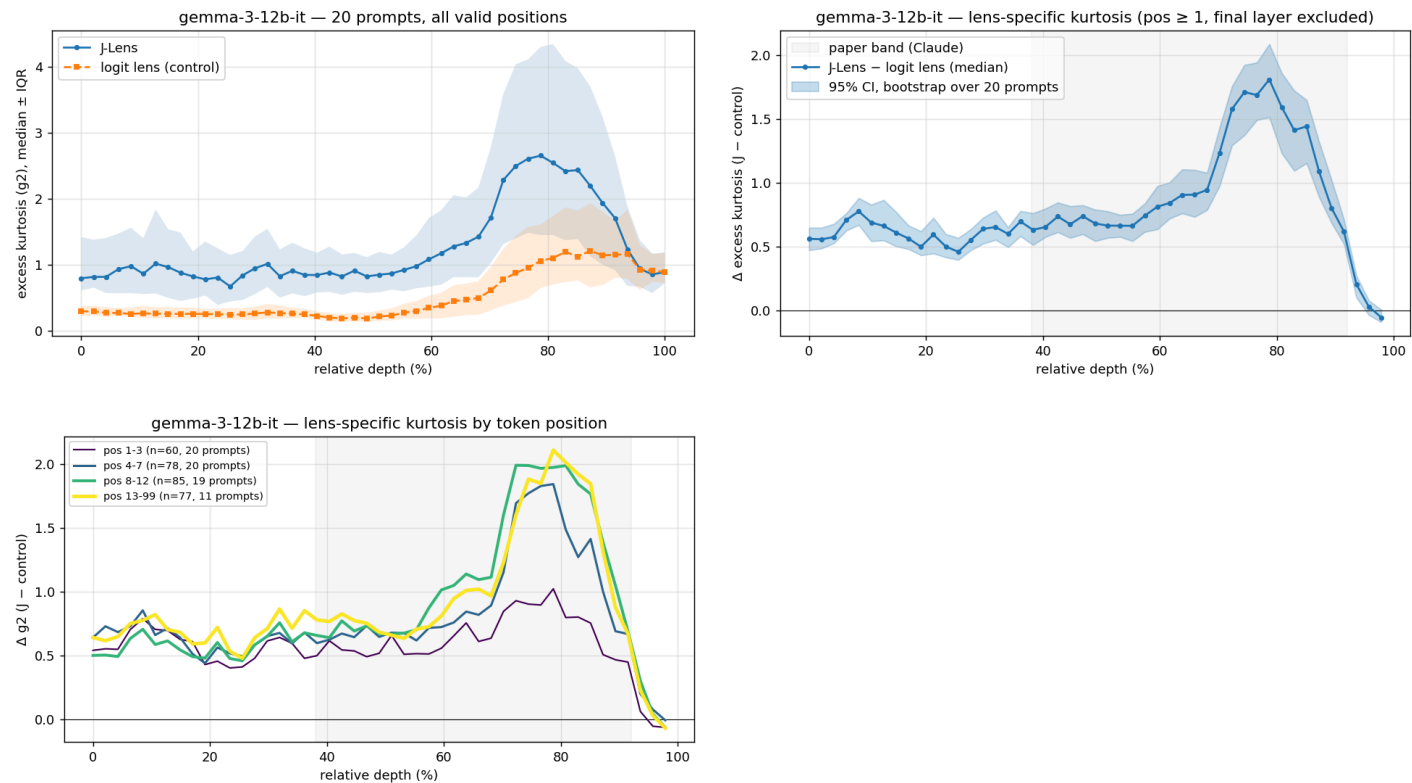
clear position effect inside the same prompts



gemma-3-12b-it 48 layers · d_model 3840 · bfloat16

J-C baseline	J-C peak	bump	CI95 (prompts)	prompts > 0	onset	pos 1-5 → 11-15
0.66	1.81 @ 79 %	1.15	[0.90 ; 1.37]	20/20	70 %	1.09 → 2.44 (10/10)

clear position effect inside the same prompts

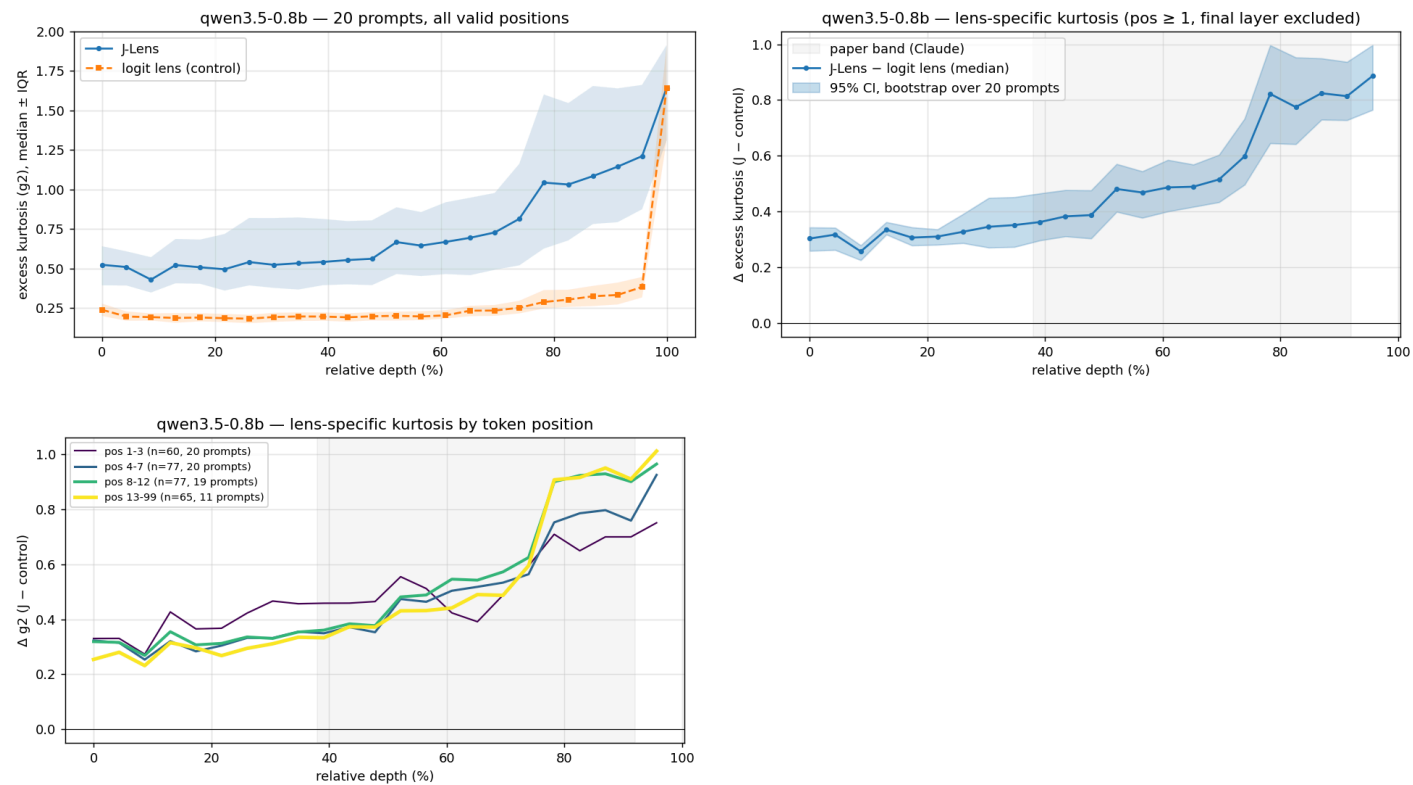


Family qwen3.5

qwen3.5-0.8b24 layers · d_model 1024 · bfloat16

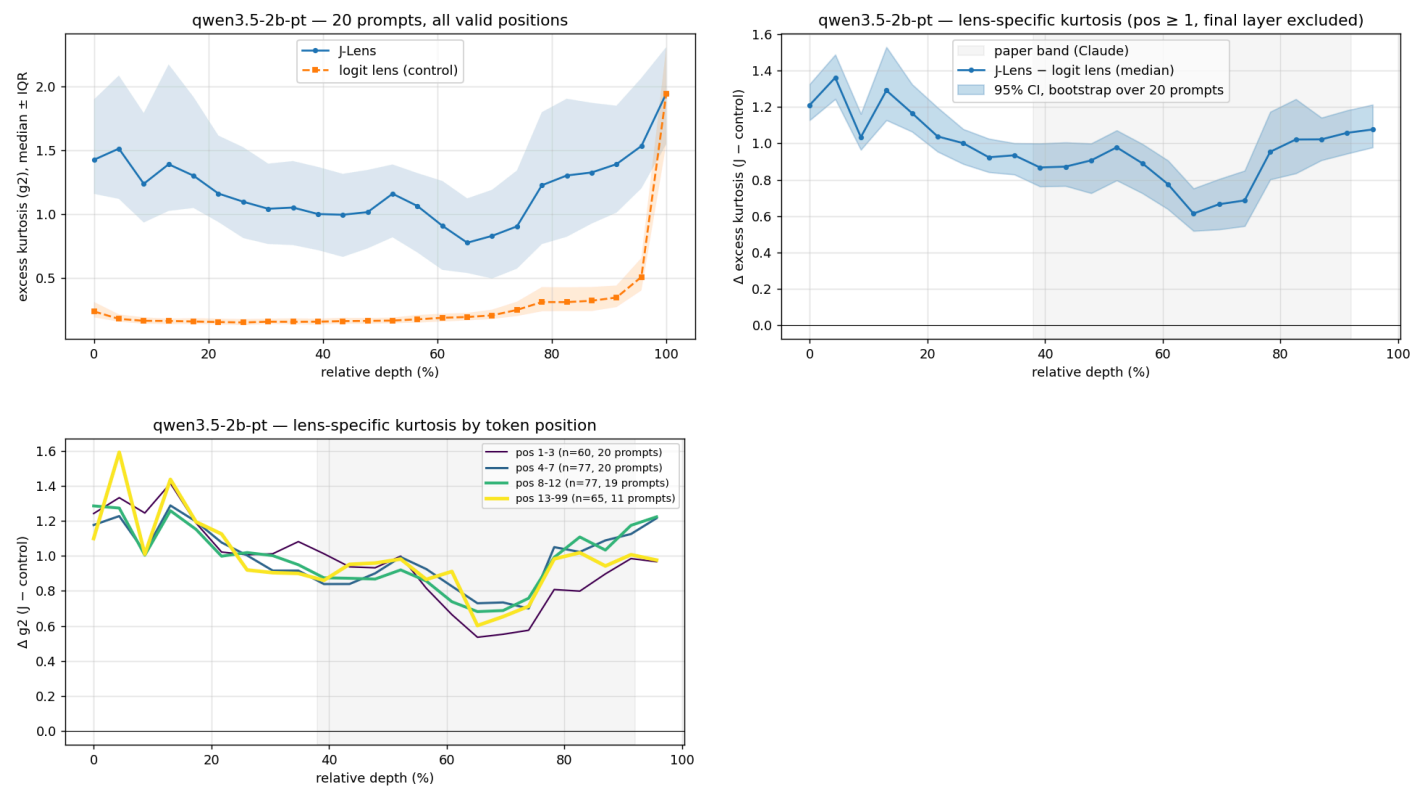
J-C baseline	J-C peak	bump	CI95 (prompts)	prompts > 0	onset	pos 1-5 → 11-15
0.36	0.82 @ 87 %	0.46	[0.39 ; 0.58]	20/20	78 %	0.55 → 0.89 (7/8)

clear position effect inside the same prompts



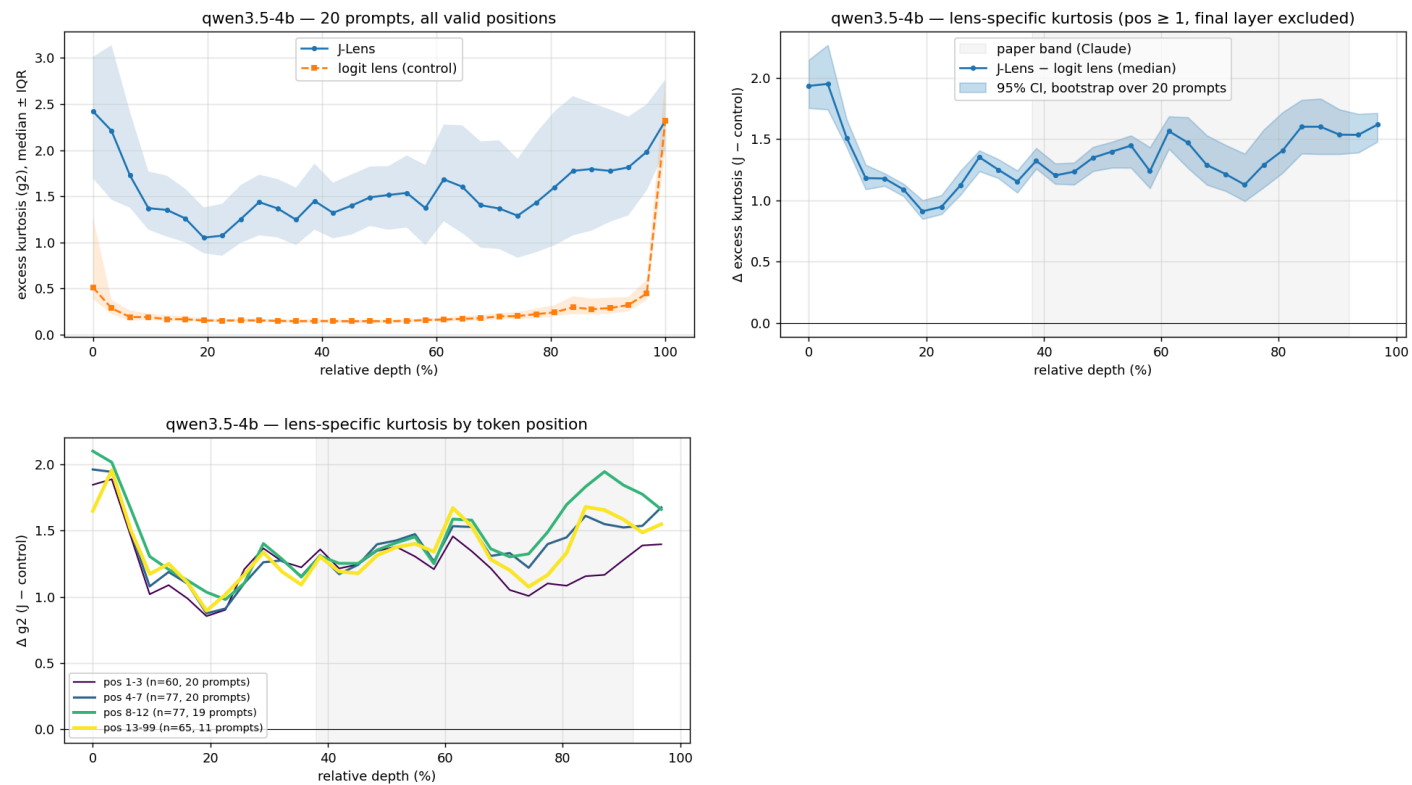
qwen3.5-2b-pt24 layers · d_model 2048 · bfloat16

J-C baseline	J-C peak	bump	CI95 (prompts)	prompts > 0	onset	pos 1-5 → 11-15
0.92	1.06 @ 91 %	0.13	[0.05 ; 0.29]	19/20	—	0.89 → 1.00 (5/8)



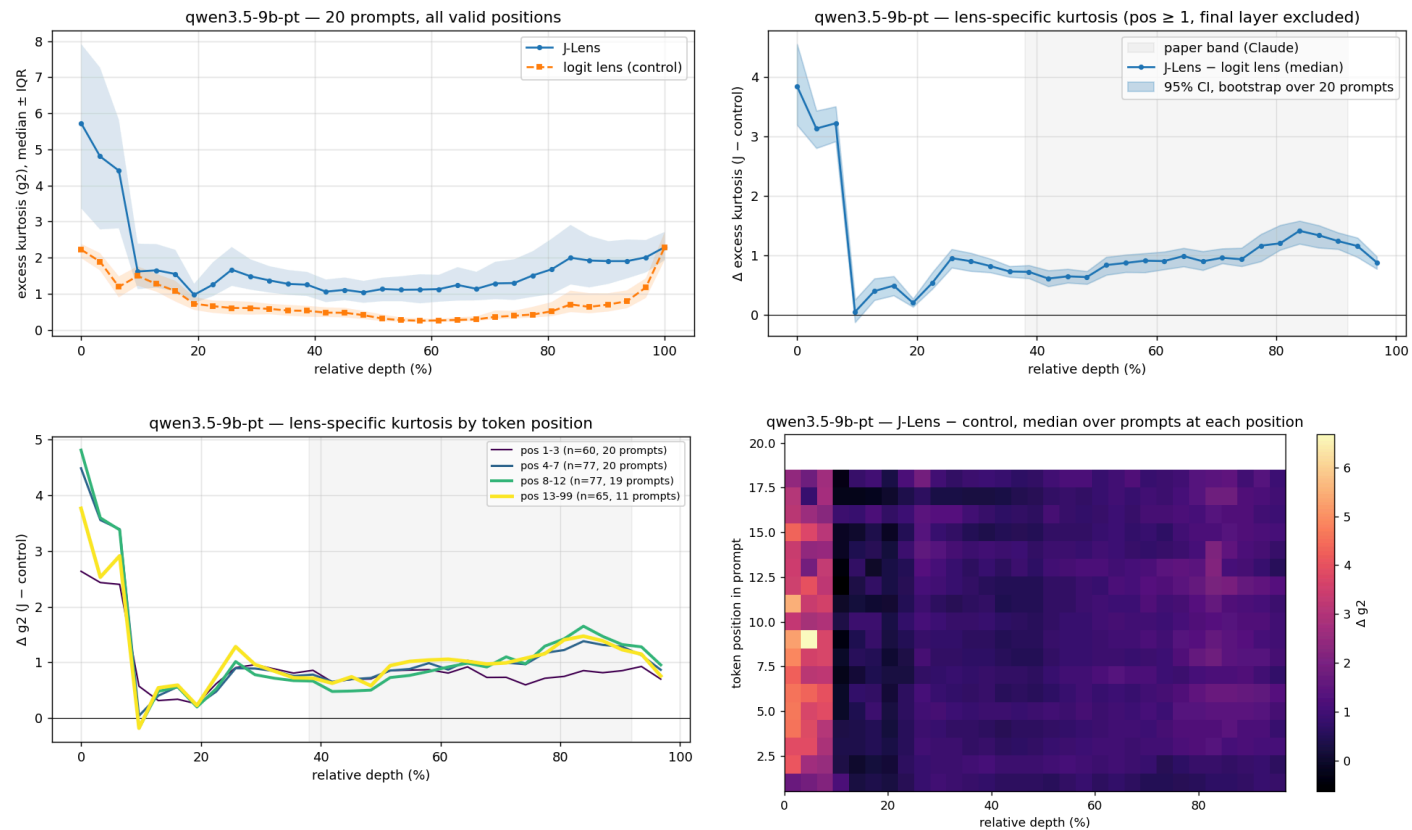
qwen3.5-4b 32 layers · d_model 2560 · bfloat16

J-C baseline	J-C peak	bump	CI95 (prompts)	prompts > 0	onset	pos 1-5 → 11-15
1.29	1.60 @ 84 %	0.31	[0.21 ; 0.52]	19/20	84 %	1.44 → 1.61 (3/8)



qwen3.5-9b-pt 32 layers · d_model 4096 · bfloat16

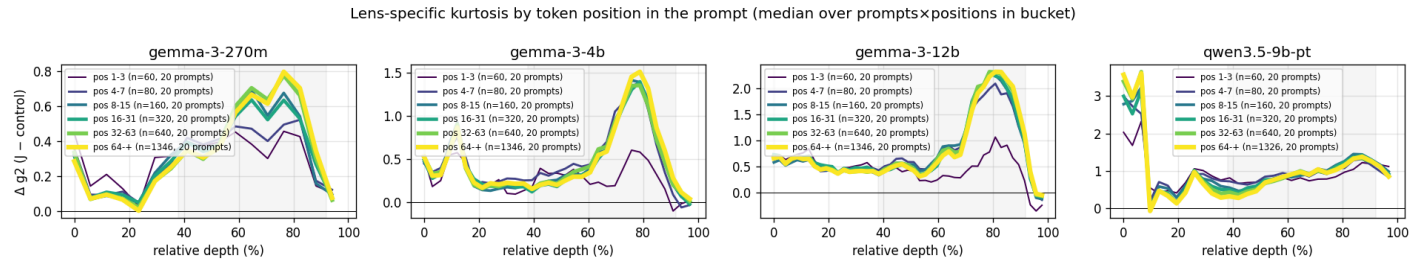
J-C baseline	J-C peak	bump	CI95 (prompts)	prompts > 0	onset	pos 1-5 → 11-15
0.77	1.41 @ 84 %	0.64	[0.44 ; 0.82]	20/20	77 %	1.00 → 1.51 (6/8)



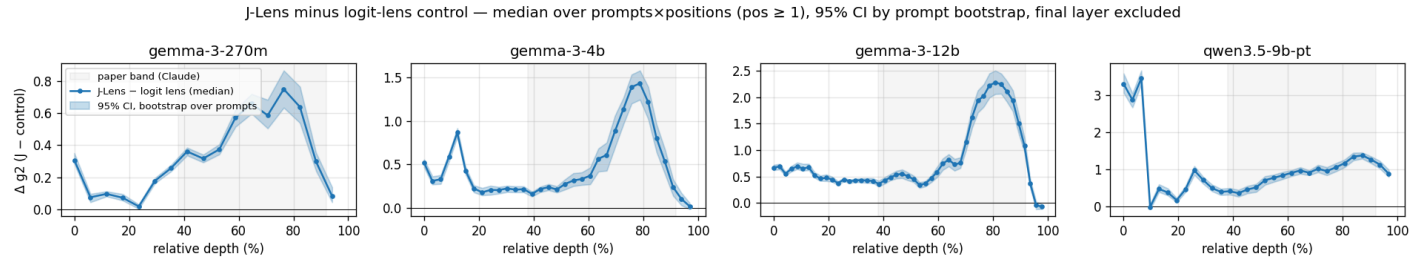
Context axis — long-prompt run

20 prompts of 110-158 tokens (prompts_long.txt), same arms, same estimator; positions ≥ 1, final layer excluded. Models: gemma-3-270m, gemma-3-4b, gemma-3-12b, qwen3.5-9b-pt.

J-C by position bucket



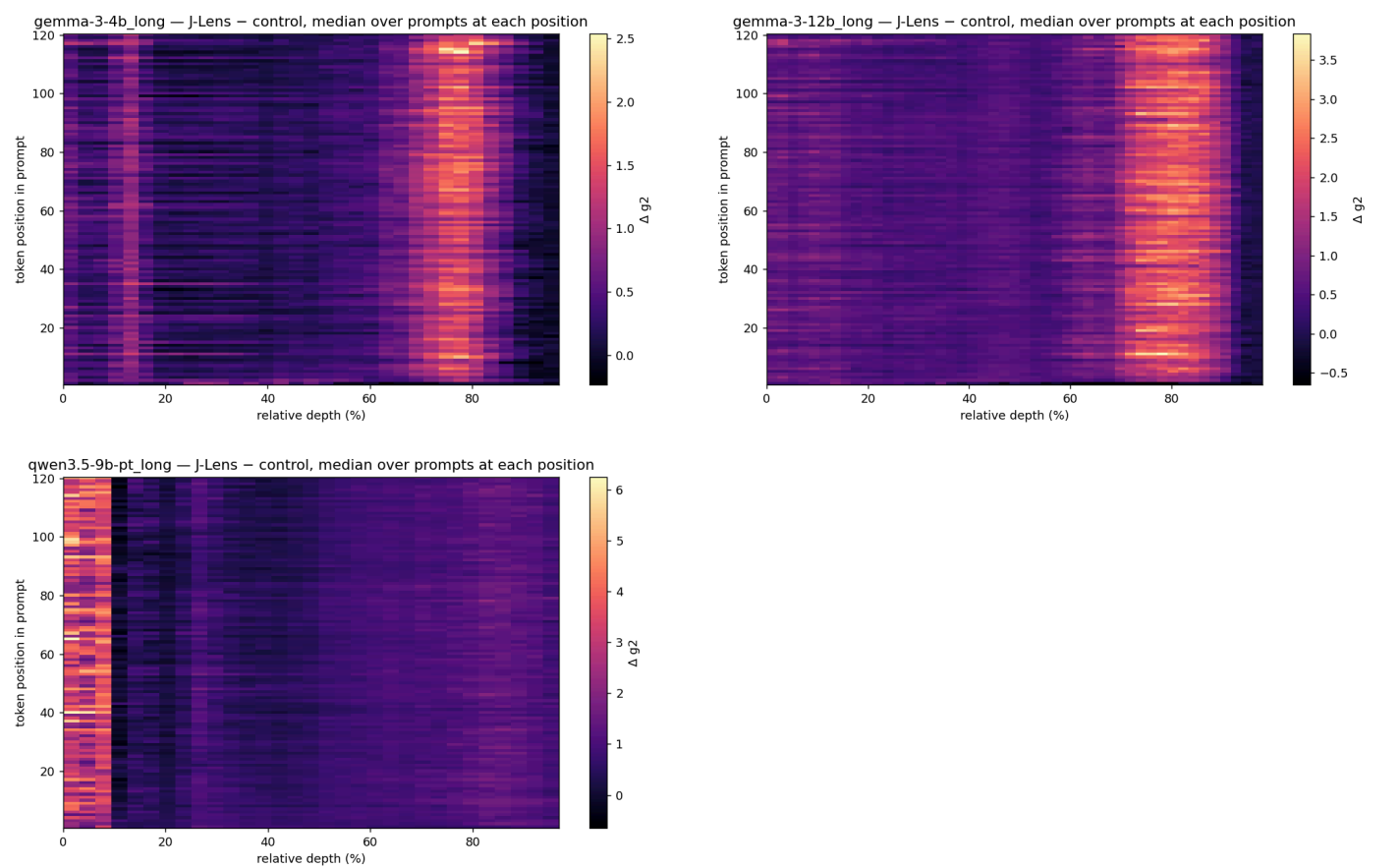
J-C with CI95 by prompt bootstrap



Within-prompt control: J-C peak at positions 1-5 vs 40-59 of the same prompts

model	prompts	peak pos 1-5	peak pos 40-59	late > early	CI95 of the gap
gemma-3-270m	20	0.51	0.76	11/20	[0.05, 0.37]
gemma-3-4b	20	0.98	1.32	10/20	[0.09, 0.60]
gemma-3-12b	20	1.32	2.26	14/20	[0.33, 1.26]
qwen3.5-9b-pt	20	1.17	1.35	12/20	[-0.01, 0.34]

Depth × position heatmaps



Summary table (by family, increasing size)

model	layers	J-C baseline	peak	bump	CI95	prompts > 0	onset	pos 1-5 → 11-15
gemma-3-270m	18	0.33	0.58 @ 65 %	0.25	[0.14 ; 0.37]	20/20	—	0.43 → 0.80 (7/10)
gemma-3-270m-it	18	0.19	0.38 @ 65 %	0.19	[0.12 ; 0.30]	20/20	—	0.27 → 0.57 (9/10)
gemma-3-1b	26	0.31	0.35 @ 68 %	0.05	[−0.23 ; 0.30]	13/20	—	0.12 → 0.50 (7/10)
gemma-3-1b-it	26	0.60	0.75 @ 60 %	0.16	[−0.00 ; 0.29]	12/20	48 %	0.92 → 0.98 (5/10)
gemma-3-4b	34	0.23	1.23 @ 79 %	0.99	[0.79 ; 1.27]	20/20	64 %	0.92 → 1.34 (8/10)
gemma-3-4b-it	34	0.41	0.81 @ 76 %	0.40	[0.24 ; 0.55]	19/20	76 %	0.51 → 1.05 (7/10)
gemma-3-12b	48	0.51	2.37 @ 79 %	1.87	[1.58 ; 2.17]	20/20	62 %	1.51 → 2.64 (9/10)
gemma-3-12b-it	48	0.66	1.81 @ 79 %	1.15	[0.90 ; 1.37]	20/20	70 %	1.09 → 2.44 (10/10)
qwen3.5-0.8b	24	0.36	0.82 @ 87 %	0.46	[0.39 ; 0.58]	20/20	78 %	0.55 → 0.89 (7/8)
qwen3.5-2b-pt	24	0.92	1.06 @ 91 %	0.13	[0.05 ; 0.29]	19/20	—	0.89 → 1.00 (5/8)
qwen3.5-4b	32	1.29	1.60 @ 84 %	0.31	[0.21 ; 0.52]	19/20	84 %	1.44 → 1.61 (3/8)
qwen3.5-9b-pt	32	0.77	1.41 @ 84 %	0.64	[0.44 ; 0.82]	20/20	77 %	1.00 → 1.51 (6/8)